CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 8: WRITE THE PYTHON POGRAM TO IMPLEMENT DFS

Aim:

To write a Python program to implement Depth-First Search (DFS) traversal of a graph.

Algorithm:

1. Start the program.
2. Represent the graph using an adjacency list.
3. Initialize an empty set to store visited nodes.
4. Start DFS from the given starting node.
5. Mark the current node as visited and display it.
6. Visit each unvisited adjacent node recursively.
7. Repeat until all reachable nodes are visited.
8. Stop the program.

PY8.py - C:/Users/navit/PY8.py (3.12.10)

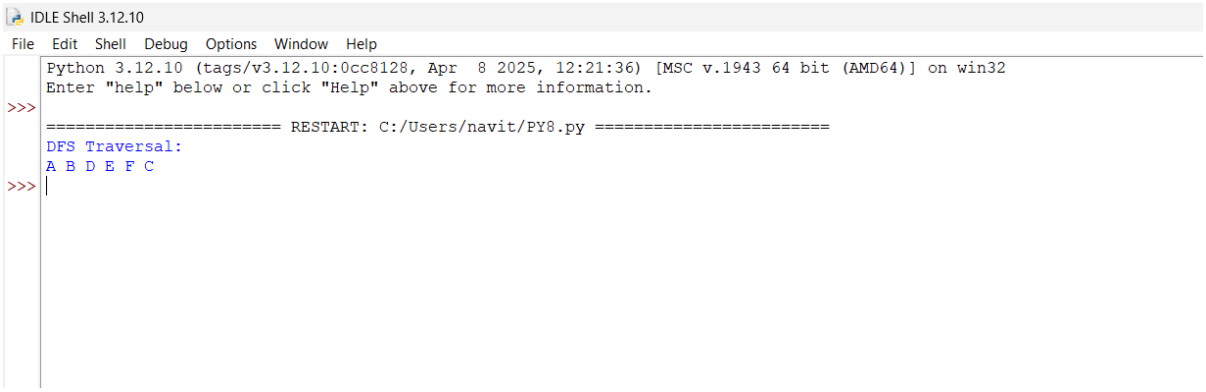
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```
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)


graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'F'],
    'D': ['B'],
    'E': ['B', 'F'],
    'F': ['C', 'E']
}

print("DFS Traversal:")
visited = set()
dfs(graph, 'A', visited)
```



Result:

The Python program was executed successfully, and the Depth-First Search traversal was obtained.